

Supplemental Material: Excluded Records and Studies at the Full-Text Stage

1. Overview

After title and abstract screening, 124 records were selected for full-text retrieval. Of these, **7 reports could not be retrieved in full text**, leaving 117 full-text studies for eligibility and Quality Assessment (QA). Following the QA assessment, 110 studies met the inclusion criteria for the final synthesis, **while 7 studies were excluded**. This document provides a transparent record of both: (1) reports not retrieved and (2) full-text studies excluded after eligibility and QA assessment.

#	Author (Year)	Paper Title	Reason for Exclusion
1	Agarkar (2025)	Assessing the impact of user behavior and insider threats on critical infrastructure	Lacks technical detail and specific algorithms for the data preprocessing stage (cleaning, normalization).
2	Ahmad (2021)	Mitigating malicious insider attacks in the internet of things using Supervised machine learning techniques	Utilizes the NSL-KDD network dataset; reflects low-level network traffic instead of organizational user behavior.
3	Alnaim (2025)	Zero-Trust Mechanisms for Securing Distributed Edge and Fog Computing in 6G Networks	Focuses on network traffic/6G hardware trust rather than human behavioral feature engineering.
4	Kandanaarachchi (2021)	Honeyboost: Boosting honeypot performance with data fusion and anomaly detection	Focused on network protocols and honeypots without expert-defined human behavioral indicators.
5	Mahmoudi-Nasr (2024)	An authorization framework to mitigate insider threat in CIM-based smart grid	Utilizes manual statistical formulas (EWMA) rather than ML/DL-based feature engineering frameworks.
6	Sulaiman et al. (2023)	Quantitative Analysis of Worm Transmission and Insider Risks in Air-Gapped Networking Using a Novel Machine Learning Approach	Uses ML as a mathematical solver for worm transmission equations rather than an ITD detection framework.
7	Yusof et al. (2022)	Behavioral Intrusion Prediction Model on Bayesian Network over Healthcare Infrastructure	Relies strictly on network-layer features (PCAP) and lacks human/expert behavioral rules.

Table 2: Detailed Quality Assessment (QA) Scoring for Excluded Studies

The following papers were evaluated against five Quality Assessment criteria (QA1–QA5). Each criterion is scored out of 1.0, with a maximum possible score of 5.0.

QA Criteria Legend:

- **QA1:** Dataset Validity (Specific to ITD)
- **QA2:** Preprocessing Rigor (Cleaning/Normalization)
- **QA3:** Feature Engineering/Expert Indicators
- **QA4:** Model Architecture Depth
- **QA5:** Performance Metrics & Evaluation

#	Author / Study	QA1: Dataset/Data Source	QA2: Dataset Characteristics	QA3: Preprocessing	QA4: Evaluation Setup	QA5: Limitations/Reproducibility	Total	Decision
1	Agarkar (2025), <i>Assessing user behavior and insider threats in critical infrastructure</i>	1.0: Logs/interviews identified	0.5: Size, labels, scenarios unclear	0.0: No cleaning/normalization detail	0.5: Results reported; validation unclear	0.0: No access/reproducibility detail	2.0	Exclude
2	Ahmad (2021), <i>Mitigating malicious insider attacks in IoT using supervised ML</i>	1.0: NSL-KDD identified	0.5: Network-flow data; weak UBA fit	0.5: General normalization only	1.0: Comparative ML evaluation reported	0.0: Limited scope/reproducibility discussion	3.0	Exclude

3	Alnaim (2025), <i>Zero-trust mechanisms for edge/fog 6G networks</i>	0.5: Network/security dataset used	0.5: Lacks insider-user context	0.5: General preparation only	1.0: Security framework evaluated	0.0: Limited ITD dataset limitations	2.5	Exclude
4	Kandanaarachchi (2021), <i>Honeyboost: Honeypot performance with data fusion</i>	0.5: LAN monitoring data used	0.5: Protocol data; no insider labels	0.5: Fusion/anomaly prep only	0.5: Honeypot, not ITD evaluation	0.0: Limited ITD reproducibility detail	2.0	Exclude
5	Mahmoudi-Nasr (2024), <i>Authorization framework for smart-grid insider threat</i>	0.5: Private smart-grid data	0.5: Limited size/label/scenario detail	0.5: CIM/statistical preparation only	0.5: Authorization, not ML-ITD evaluation	0.0: Limited access/reproducibility detail	2.0	Exclude
6	Sulaiman et al. (2023), <i>Worm transmission and insider risks in air-gapped networks</i>	0.0: No behavioral ITD dataset	0.0: No relevant dataset details	0.0: No behavioral preprocessing	0.5: Mathematical evaluation only	0.0: No ITD reproducibility evidence	0.5	Exclude
7	Yusof et al. (2022), <i>Behavioral intrusion prediction in healthcare infrastructure</i>	1.0: Healthcare PCAP data used	0.5: Traffic-level details only	0.5: Feature selection; limited preprocessing	1.0: Bayesian evaluation reported	0.0: Limited transferability/reproducibility detail	3.0	Exclude

The following 7 records were selected for full-text retrieval but could not be downloaded or accessed in full text. Therefore, they were not included in the 117 studies assessed for eligibility and QA.

1- Insider threat detection and prevention using semantic score and dynamic multi-fuzzy classifier Malvika Singh , S. Sangeetha and B.M. Mehtre 2023

2- Explainable AI (XAI) for Insider Threat Detection: Balancing Security and Transparency in Cloud Computing Arul Selvam P, Tamije Selvy P 2025

3- User behaviour based insider threat detection model using an LSTM integrated RF model
S. K. Uma Maheswaran,L. Rajasekar,Ziaul Haque Choudhury &Makarand Shahade 2023

4- Identification of authorship and prevention of fraudulent transactions/cybercrime using efficient high performance machine learning techniques

B.J. Sowmya, R. Hanumantharaju, D. Pradeep Kumar and K.G. Srinivasa 2022

5- Deep Temporal Graph Infomax for Imbalanced Insider Threat Detection
Peng Gao Icon,Haotian Zhang,Ming Wang,Weiyong Yang,Xinshen Wei,Zhuo Lv & 2023

6- Social network malicious insider detection using time-based trust evaluation T Nathezhtha, D Sangeetha, V Vaidehi 2023

7- ACGAN: Adaptive Loss–Guided GAN–CNN Framework for Semantically Coherent Synthetic Insider Threat Data Generation A Slim, M Malah 2025